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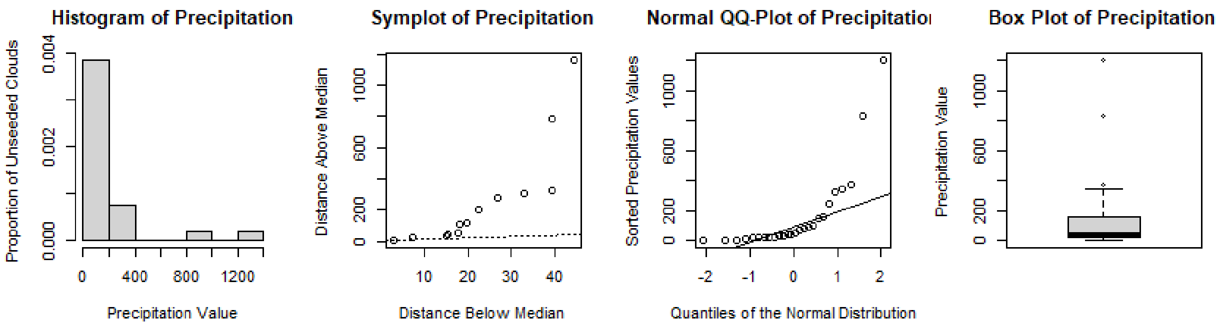
Assignment 5

Exercise 5.2

Part a.

The function *summary()* gives that the mean and median of the precipitation values of the ununseeded clouds are 165 and 44, respectively. Since the mean is greater than the median, the distribution of the precipitation values seems to be positively skewed. The plots below show a long right tail, mass concentration on the left, and a few outliers to the right of the median. This further supports the conclusion of positive skewness of the precipitation values.



Part c.

The empirical bootstrap seems to be a good choice since there are no clear assumptions about the distribution of the precipitation values. Hence, we draw 2000 bootstrap samples from the given precipitation data of unseeded clouds and calculate corresponding standard deviation estimates. The bootstrap estimate of the standard deviation of the standard deviation estimator is approximately 83. The 2000 replicates seem to sufficiently reduce the influence of randomness on the bootstrap estimate of the standard deviation.

Part d.

The procedure above with the MAD estimator instead of the standard deviation estimator gives that the bootstrap estimate of the standard deviation of the MAD estimator is approximately 32.

Part e.

It follows from Part c. and d. that the standard deviation of the standard deviation estimator is greater than that of the MAD estimator. This means that the standard deviation estimator deviates more around its mean compared to the MAD estimator. This occurs because the standard deviation estimator measures deviation of a drawn bootstrap sample around its mean, but the MAD estimator measures deviation of the drawn bootstrap sample around its median. Since some outliers are present in the given data, as it can be seen in the box plot in Part a., the values of the standard deviation estimator deviate more than the values of the MAD estimator. Hence, we prefer MAD as a more robust estimator for accuracy because it damps down the effect of outliers.

Part f.

It follows from Part a. that the distribution of the precipitation value seems to be positively skewed. This suggests that the distribution is asymmetrical and not normally distributed because the normal distribution is symmetric. Hence, the t-test cannot be used because of the normality assumption, and the signed rank test cannot be used because of the symmetricity assumption. Therefore, we prefer the sign test.

Part g.

The sign test assumes that there is a unique median, and there is no observation equal to 40. Indeed, there is no such observation in the data. The function *binom.test()* yields p-value 0.42. This p-value is not lower than 0.05; hence, we cannot reject the null hypothesis stating that the median is lower than 40. The relatively big value of the p-value suggests that the null hypothesis is true.

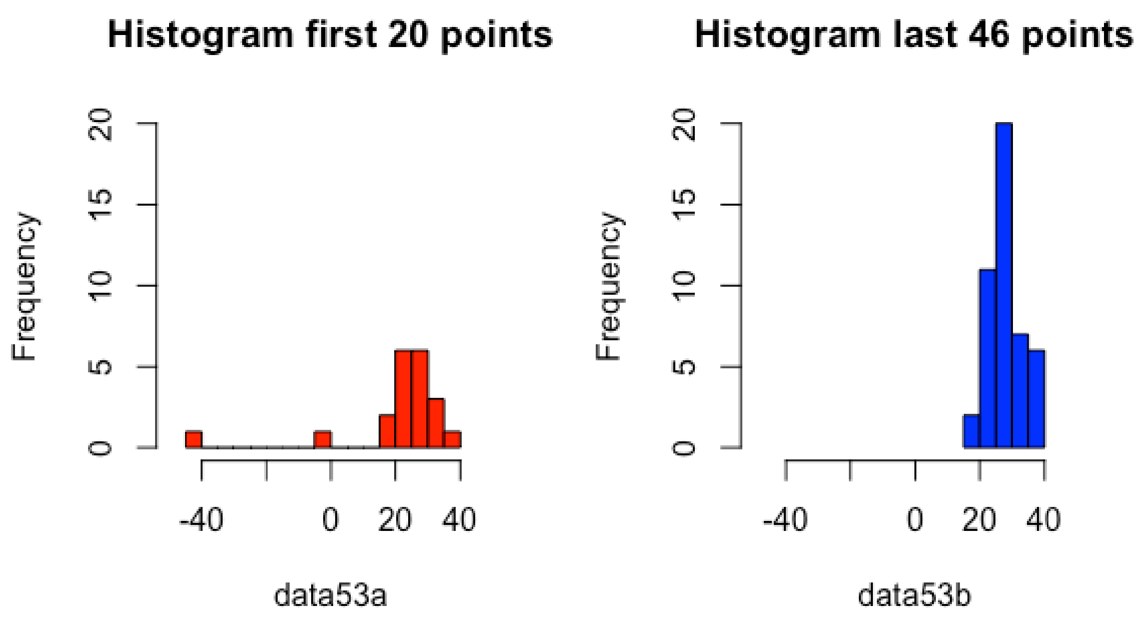
Part i.

The sign test seems to give the most valuable confidence interval because the provided data satisfy assumptions of the sign test better than they satisfy the (symmetricity) assumptions of the signed rank test and the t-test, and the confidence interval of the sign test is tighter compared to the confidence intervals provided by the signed rank test and the t-test which allows us to reject more location parameters.

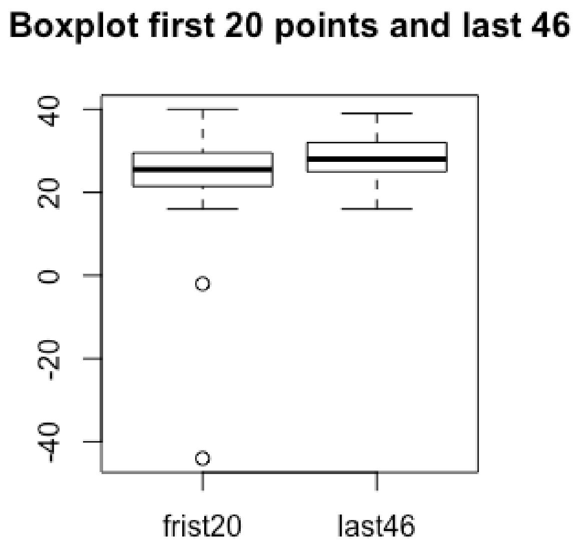
Exercise 5.3

Part a.

We are looking at the differences between the first 20 data points and the rest of the data points. In total there are 66 data points. The graphical summaries show that the median may lower for the first 20 points. The most notable difference is the 2 outliers in the 20 data points. Both outliers are below the mean. Above the average, it is notable that it is also high top value but this is not seen as an outlier.



In the later 46 point the data is extremely focused around 25 and 30, while this is less apparent in the first 20 points.



The data is more focused around the median by the 46 data points.

Two suitable tests to test if the data set are different are Kolmogorov-Smirnov test and Mann-Whitney test which find out if the two sample are from the same distribution. The significance level we choose is 5%.

Hypotheses

H_0: D1 = D2
H_1: D1 $\neq$ D2

where D1 is the distribution of the first 20 points and D2 is the distribution of the last 46 points.

The ks.test gives a p-value of 0.328 so we reject the null hypothesis. The wilcox.test gives us a p-value of 0.1189 so we keep rejecting the null hypothesis. So, from the test and graphical summaries we conclude that there is not a significant difference between the two distributions of the samples.

Part b.

To test if the True value is 24.8332, we test with a wilcox.test with a confidence interval of 0.95.

H_0: TrueValue (TV) = 24.8332
H_1: TV $\neq$ 24.8332

The p-value is 0.00029 and the given interval is [26,28.5]. The interval does not match the H_0 and the p-value is lower tha0.05. So there for the finding from this old experiment are out dated form the physic we now know.

Appendix

```
setwd("C:/Users/david/OneDrive/VU/Statistical Data Analysis/Assignment 5")
source('functions_Ch3.txt')
source('functions_Ch5.txt')
source('clouds.txt')
source('statgrades.txt')

# Exercise 5.1
data51 <- scan("statgrades.txt")
hist(data51)
summary(data51)

 # Part a.
g62 = 0
for(i in 1:length(data51)){
 if(data51[i] > 6.2){
  g62 = g62 + 1
 }
}

binom.test(g62,length(data51),alternative="less")
pval_a = 0.001444
reject_a = TRUE

 # Part b.
data51b = NULL
for(i in 1:length(data51)){
 if(data51[i] != 6.0){
  data51b[length(data51b)+1] = data51[i]
 }
}

g6 = 0
for(i in 1:length(data51b)){
 if(data51b[i] > 6.0){
  g6 = g6 + 1
 }
}

binom.test(g6,length(data51b), alternative = "two.sided")
pval_b = 0.211
reject_b = FALSE

 # Part c.
data51c = NULL
for(i in 1:length(data51)){
 if(data51[i] != 5.5){
```

```
   data51c[length(data51c)+1] = data51[i]
  }
}

g55 = 0
for(i in 1:length(data51c)){
  if(data51c[i] >= 5.5){
    g55 = g55 + 1
  }
}

binom.test(g55,length(data51c),p = 0.45, alternative = "g")
pval_c = 0.0568
reject_c = TRUE

# Exercise 5.2

  # Part a.
clouds = read.table('clouds.txt')
summary(clouds$unseeded.clouds)

par(mfrow=c(1,4), pty="s")
hist(clouds$unseeded.clouds, prob=T, main="Histogram of Precipitation",
     xlab="Precipitation Value",
     ylab="Proportion of Unseeded Clouds")
symplot(clouds$unseeded.clouds, main="Symplot of Precipitation")
qqnorm(clouds$unseeded.clouds, main="Normal QQ-Plot of Precipitation",
       xlab="Quantiles of the Normal Distribution",
       ylab="Sorted Precipitation Values")
qqline(clouds$unseeded.clouds)
boxplot(clouds$unseeded.clouds, main="Box Plot of Precipitation",
        ylab="Precipitation Value")

  # Part b.
sd(clouds$unseeded.clouds)

  # Part c.
sd(bootstrap(clouds$unseeded.clouds, sd, B = 2000))

  # Part d.
sd(bootstrap(clouds$unseeded.clouds, mad, B = 2000))

  # Part g.
c = sum(clouds$unseeded.clouds == 40)
x = sum(clouds$unseeded.clouds > 40)
n = length(clouds$unseeded.clouds)

binom.test(x,n-c,alt="g")
```

```
  # Part h.
t.test(clouds$unseeded.clouds)
wilcox.test(clouds$unseeded.clouds, conf.int = T)

rbind(0:26, round(pbinom(0:26,size=26,p=0.5),3))
sort(clouds$unseeded.clouds)

1 - pbinom(16,25,0.5)
pbinom(8,25,0.5)

# Exercise 5.3

  # Part a.
data53 = scan('newcomb.txt')
data53a = data53[1:20]
data53b = data53[21:66]
summary(data53a)
summary(data53b)
par(mfrow=c(1,1), pty="s")
boxplot(data53a, data53b, main = "Boxplot first 20 points and last 46", names = c('frist20','last46'))
par(mfrow=c(1,2), pty="s")
hist(data53a, xlim = c(-50,50), ylim = c(0,20), col = 'red', nclass = 20, main = "Histogram first 20
points")
hist(data53b, xlim = c(-50,50), ylim = c(0,20), col = 'blue', main = "Histogram last 46 points")
ks.test(data53a, data53b, alternative = "two.sided")
wilcox.test(data53a,data53b ,alternative = "two.sided")

  # Part b.
wilcox.test(data53, mu = 24.8332, conf.int = 0.05)

mylist <- list(stud_no = c(2669137 , 2671011),
            "1_a_pval" = pval_a, '1_a_reject' = reject_a,"1_b_pval" = pval_b,
            '1_b_reject' = reject_b,"1_c_pval" = pval_c, '1_c_reject' = reject_c,
            "2_b_sd" = sd(clouds$unseeded.clouds),
            "2_d_mad" = sd(bootstrap(clouds$unseeded.clouds, mad, B = 2000)),
            "2_h_CI_sign" = c(26.3 , 147.8),
            "2_h_CI_wilcox" = c(36.69997 , 187.24999),
            "2_h_CI_t" = c(52.09509 , 277.02876)
            )

save(mylist, file="C:/Users/david/OneDrive/VU/Statistical Data Analysis/Assignment
5/myfile5_39`.RData")
```